

SCHOOL OF SCIENCE AND ENGINEERING
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING



EGE320 Electronics 1 – Fall 2023

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COURSE PROJECT: Design of a Two-Stage Amplifier

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Introduction

The objective of this project was to use 2N2222 NPN bipolar transistors to design an amplifier with a gain of around -13V/V when the circuit was loaded with a 0.5kΩ resistor. The fluctuation of the voltage across the emitter-base junction (v_{be}) was limited to 5mV to avoid large signal distortion. Two configurations were considered for the implementation. One was a single-stage inverting amplifier using a common emitter amplifier (CE). The other was a two-stage inverting amplifier using a common emitter amplifier (CE) and a common collector amplifier (CC). The circuits of CE and CC amplifiers are shown in **Figure 1**.

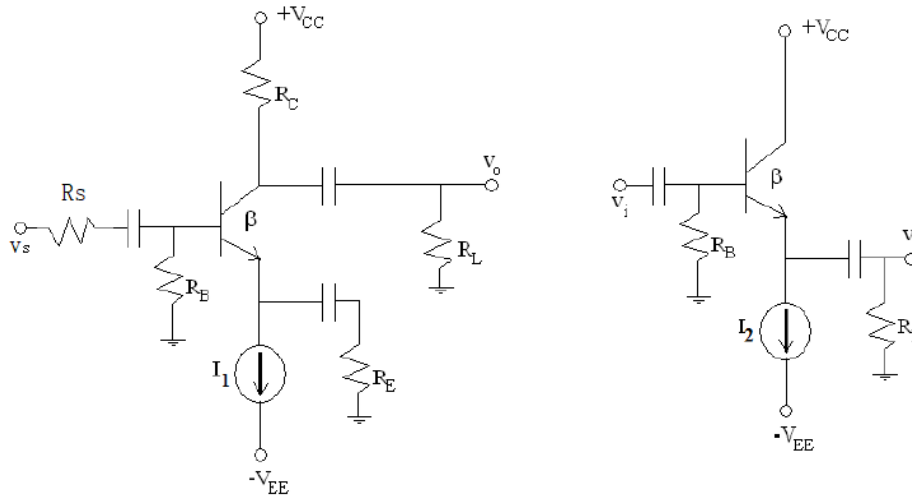


Figure 1. CE and CC Amplifiers

The available components and provided data are as follows.

1. Signal source: $v_s(t) = 50[mV]\sin(2\pi 10^3 t)$ with an internal resistance $R_s = 50[\Omega]$.
2. Two 2N2222 NPN bipolar transistors with the current gain $\beta = 200$.
3. Operating temperature: room temperature, therefore $\eta V_T \approx 25[mV]$.
4. Power supplies: $V_{CC} = 10[V]$, $-V_{EE} = -10[V]$, and two current sources whose current I can be adjusted to any value between $1[mA] \leq I \leq 2[mA]$.
5. Resistors: $R_B = 20[k\Omega]$, $R_C = 5[k\Omega]$, $R_L = 0.5[k\Omega]$.
6. Capacitors: $100\mu F$.

The design started with biasing current $I_1 = I_2 = 1[mA]$.

Analysis

DC Analysis

The equivalent DC circuit of CE and CC Amplifiers are shown in **Figure 2**. Since the capacitors do not allow the flow of direct current, they act as an open circuit and therefore are removed. The purpose for the DC analysis is to ensure that the transistors at their respective Q points are in the active mode. The circuits show that the base and the emitter of both transistors are indirectly connected to GND and -10[V], respectively. So, both of the emitter-base junctions (EBJ) are forward biased. The voltage at the emitter V_E is

$$V_E = V_B - 0.7 = 0 - R_B I_B - 0.7 = -R_B I_E / (\beta + 1) - 0.7 = -20/201 - 0.7 = -0.8[V]$$

For the CE circuit, the voltage at the collector $V_C = V_{CC} - R_C I_C \approx 10 - 5 \times 1 = 5[V]$. So,

$V_{CE} = V_C - V_E = 5.80[V] > 0.2[V]$. For the CC circuit, the voltage at the collector

$V_C = 10[V]$. So, $V_{CE} = V_C - V_E = 10.80[V] > 0.2[V]$. Thus, with the given data, the

transistors in both CE and CC amplifiers are in active mode.

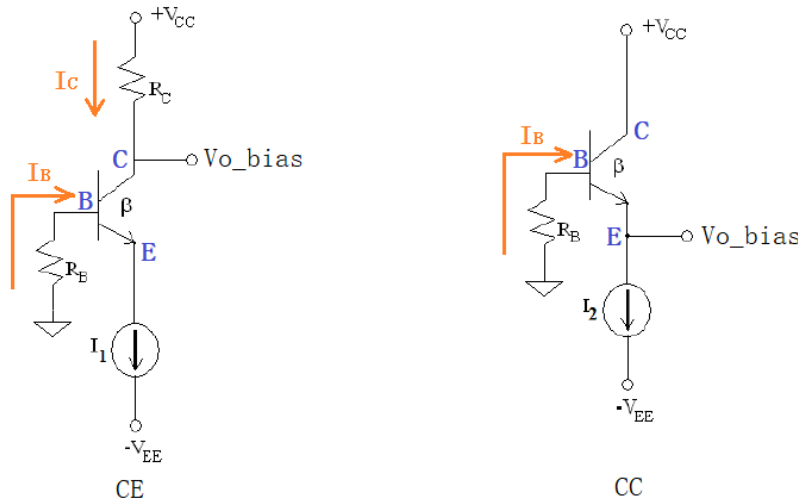


Figure 2. DC Circuits for CE and EE Amplifiers

Equivalent Model for NPN Bipolar Transistor

The circuitry for an NPN bipolar transistor is replaced with the following model by substituting the emitter-base junction (EBJ) with a resistor r_E and inserting a dependent current source to represent i_C . This model is used in AC analysis assuming the magnitude of the AC input is very small.

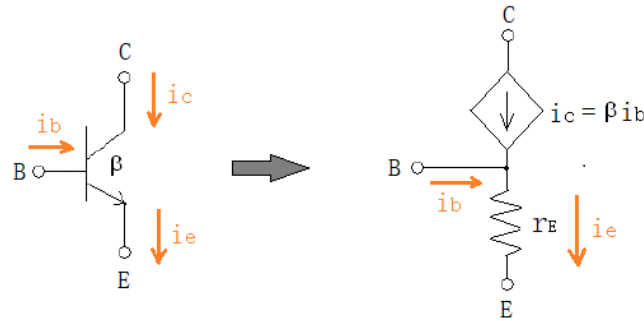


Figure 3. Equivalent Model for an NPN bipolar transistor.

The EBJ of an NPN bipolar transistor acts as a diode. Its characteristic function and curve are shown in **Figure 4** below.

$$i_E = (\beta + 1)i_B = (\beta + 1)I_s e^{v_{BE}/(\eta V_T)}$$

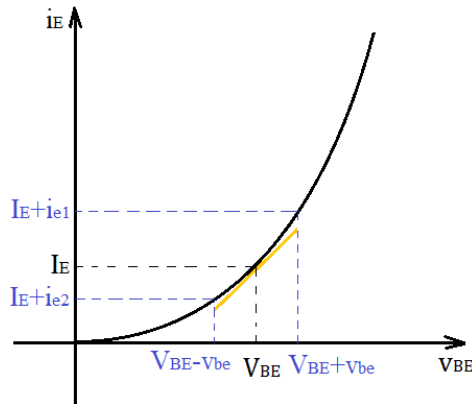


Figure 4. Characteristic Curve of EBJ

When the AC input voltage, v_{be} , is very small (usually $\leq 5[mV]$), the curve in the interval

$(V_{BE} - v_{be}, V_{BE} + v_{be})$ can be approximated as the tangent line at the Q point (V_{BE}, I_E) . So, the

current change (i_e) induced by v_{be} is now linear to v_{be} . This is why the EBJ is approximated as a resistor r_E in the model. Its resistance is calculated as follows.

$$\begin{aligned}\frac{di_E}{dv_{BE}} &= \frac{1}{\eta V_T} (\beta + 1) I_s e^{v_{BE}/(\eta V_T)} \\ \left. \frac{di_E}{dv_{BE}} \right|_Q &= \frac{1}{\eta V_T} (\beta + 1) I_s e^{V_{BE}/(\eta V_T)} = \frac{1}{\eta V_T} I_E \\ r_E &= \left. \frac{dv_{BE}}{di_E} \right|_Q = \frac{\eta V_T}{I_E} \approx \frac{25[mV]}{1[mA]} = 25[\Omega]\end{aligned}$$

If v_{be} is large, this approximation can not be applied. Otherwise, large signal distortion will occur.

AC Analysis

The simplified AC circuits for the two amplifiers are shown in **Figure 5** below. The capacitors are treated as wires because their reactance, $r = \frac{1}{\omega C} = \frac{1}{2\pi 10^3 \times 100 \times 10^{-6}} [\Omega] = 1.6[\Omega]$, is so small compared to other resistors that they can be omitted. All the DC voltage sources are shorted and the DC current sources are treated as an open circuit.

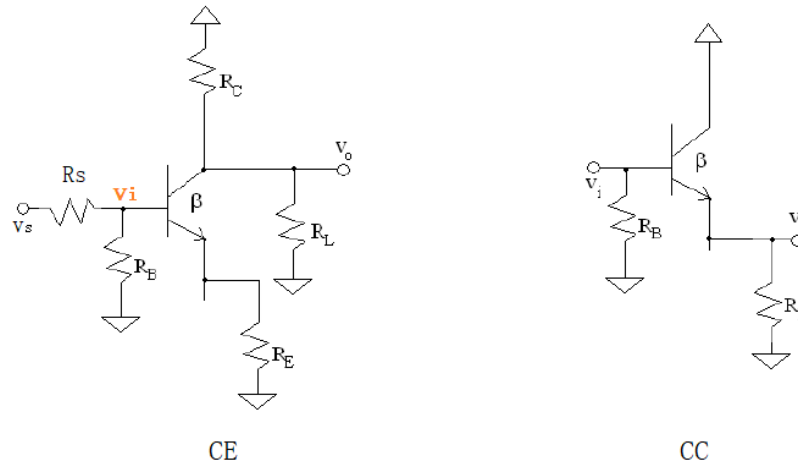


Figure 5. AC Circuits for CE and CC Amplifiers

Their equivalent circuits using the model mentioned above are shown in **Figure 6**.

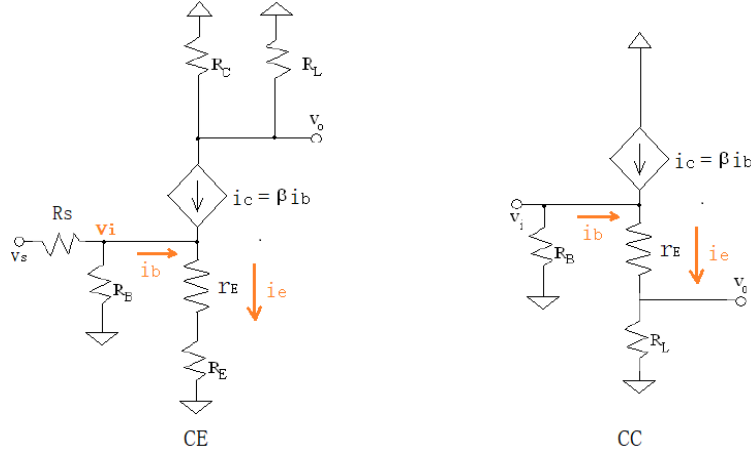


Figure 6. Models for the AC circuits of CE and CC Amplifiers.

The gains, $\frac{v_o^L}{v_s}$, of each circuit are calculated as follows.

CE:

$$i_b = (v_s - v_i)/R_s - v_i/R_B$$

$$i_c = \beta i_b = -v_o^L/R_C + (-v_o^L/R_L)$$

$$i_e = (\beta + 1)i_b = v_i/(r_E + R_E)$$

$$\Rightarrow A_1 = v_o^L/v_s = -[\beta(R_C || R_L)]/\{R_s[1 + (\beta + 1)(r_E + R_E)/(R_s || R_B)]\}$$

CC:

$$A_2 = v_o^L/v_i = R_L/(r_E + R_L)$$

Two-Stage Inverting Amplifier CE-CC

The circuit of the two-stage inverting amplifier is shown in **Figure 7** below. The output voltage v_o is the input signal of the CC amplifier. The CC amplifier is the new load of the CE amplifier.

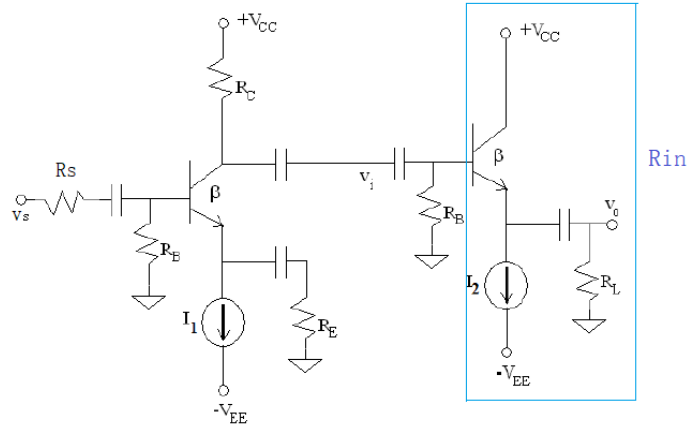


Figure 7. Two-Stage Inverting Amplifier CE-CC

First, calculate the new load resistance of the CE amplifier, R_i , which is the input equivalent resistance of the CC amplifier. From the circuit shown in **Figure 7**, we have $R_i = R_B || R_{in}$. In order to measure R_{in} , the circuit enclosed in the blue box is connected to an Ohm-meter with all its independent DC sources disabled. The simplified circuit is shown in **Figure 8** below.

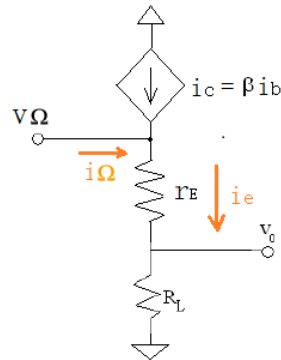


Figure 8. Circuit for the measurement of Rin

R_{in} is calculated as

$$R_{in} = v_{\Omega} / i_{\Omega} = v_{\Omega} / [i_e / (\beta + 1)] = (\beta + 1) v_{\Omega} / [v_{\Omega} / (r_E + R_L)] = (\beta + 1)(r_E + R_L)$$

So, $R_i = R_B || R_{in} = R_B || (\beta + 1)(r_E + R_L)$.

The gain of the two-stage inverting amplifier is

$$G = v_{o\ CC}^L / v_s = (v_{o\ CC}^L / v_{o\ CE}^L) (v_{o\ CE}^L / v_s) = A_2 A_1.$$

Tasks

Task 1. Attempt A Single-Stage CE Amplifier

From the Analysis section, we have

$$A_1 = v_o^L / v_s = - [\beta(R_C || R_L)] / \{ R_s [1 + (\beta + 1)(r_E + R_E) / (R_s || R_B)] \}$$

So, $-13 = - [200(5k || 0.5k)] / \{ 50 [1 + 201(25 + R_E) / 50] \}$

$$\Rightarrow R_E = 9.54[\Omega]$$

Assume $v_o^L = -13v_s = -13 \times 50[mV] = -650[mV]$, calculate v_{be} :

$$i_c = (650m / 5k + 650m / 0.5k)[A] = 1.43[mA]$$

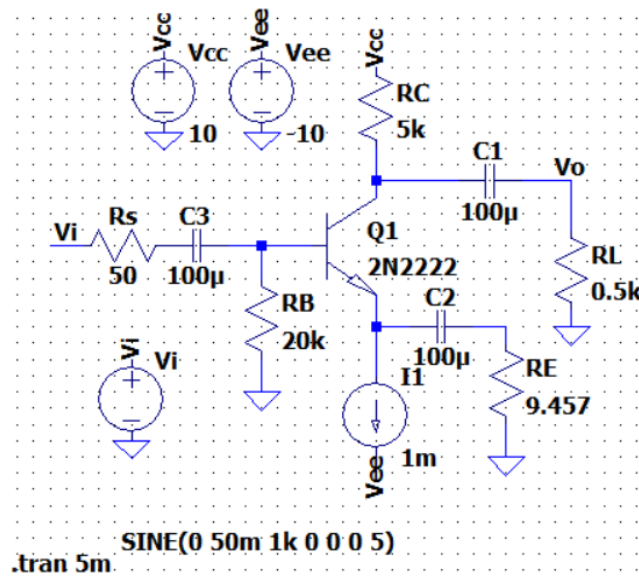
$$i_e = \frac{\beta+1}{\beta} i_c = \frac{201}{200} \times 1.43[mA] = 1.44[mA]$$

$$v_{be} = r_E i_e = 25 \times 1.44[mV] = 36[mV] > 5[mV]$$

Since v_{be} is much greater than $5[mV]$, the EBJ cannot be approximated as a resistor r_E . Thus our calculations would lead to a big distortion.

Task 2. Simulate the Single-Stage CE Amplifier

The schematic and the simulation are shown in **Figure 9** below. The simulation verified that this design did not work.



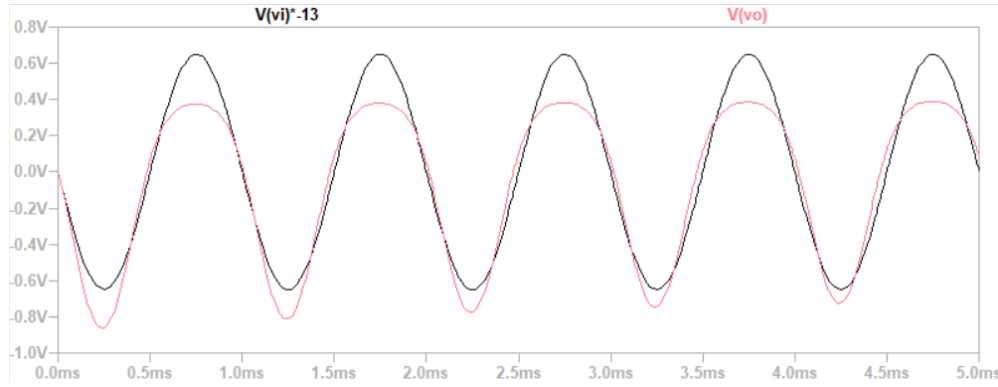


Figure 9. Simulation of the Single-Stage Inverting Amplifier

Since the value of v_{be} is large, the approximation by the BJT's model is inaccurate and the amplified input signal is not as expected.

Task 3. Attempt A Two-Stage CE-CC Amplifier

Calculate R_i :

$$R_i = R_B || R_{in} = R_B || [(\beta + 1)(r_E + R_L)] = 20k || [201 \times (25 + 0.5k)] = 16.8k[\Omega]$$

Calculate A_2 :

$$A_2 = R_L / (r_E + R_L) = 0.5k / (25 + 0.5k) = 0.952$$

Thus, $A_1 = G/A_2 = -13/0.952 = -13.65$.

Calculate R_E :

$$A_1 = -[\beta(R_C || R_i)] / \{R_s [1 + (\beta + 1)(r_E + R_E)/(R_s || R_B)]\}$$

So, $-13.65 = -[200(5k || 16.8k)] / \{50[1 + 201(25 + R_E)/50]\}$

$$\Rightarrow R_E = 255.63[\Omega].$$

Verify v_{be} :

$$i_c = (13.65 \times 50m/5k + 13.65 \times 50m/16.8k)[A] = 0.177[mA]$$

$$i_e = \frac{\beta+1}{\beta} i_c = \frac{201}{200} \times 0.177[mA] = 0.178[mA]$$

$$v_{be} = r_E i_e = 25 \times 0.178[mV] = 4.45[mV] < 5[mV]$$

Since the variation of v_{be} is within $5[mV]$, this configuration may work.

Task 4. Simulate the Two-Stage CE-CC Amplifier

The schematic and the simulation are shown in **Figure 10** below. The result showed that this design worked most of the time except when v_o dropped below $0.5[V]$. When $v_o < -0.5[V]$, $I_L = -v_o/R_L > 0.5[V]/0.5[k\Omega] = 1[mA]$. This results in $i_E = 0$ and thus the transistor Q_2 is in the cut-off mode.

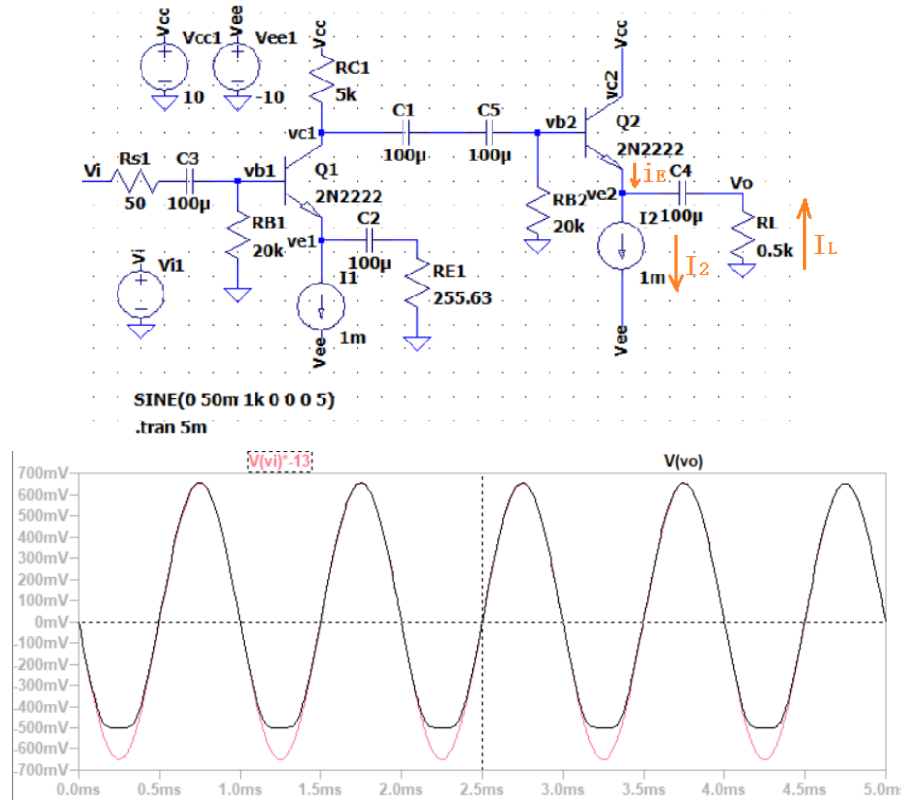


Figure 10. Simulation of the Two-Stage CE-CC Amplifier Calculated in Task 3.

Task 5. Calculate the Minimum I_2 to Avoid Cut-Off Mode of Q2.

As discussed in Task 4, the transistor Q_2 is in the cut-off mode when the output voltage v_o drops below $-0.5[V]$.

The minimum output voltage $v_o = -13v_{s-max} = -13 \times 50[mV] = -0.65[V]$. To avoid the cut-off mode, I_2 needs to be larger than I_L when $v_o = -0.65[V]$. So,

$$I_{2-min} > (0 - v_o)/R_L = 0.65[V]/0.5[k\Omega] = 1.3[mA].$$

Task 6. Simulate the Modified Two-Stage CE-CC Amplifier

The schematic and the simulation are shown in **Figure 11** below. The simulation showed that $G \approx -13$ with a small distortion.

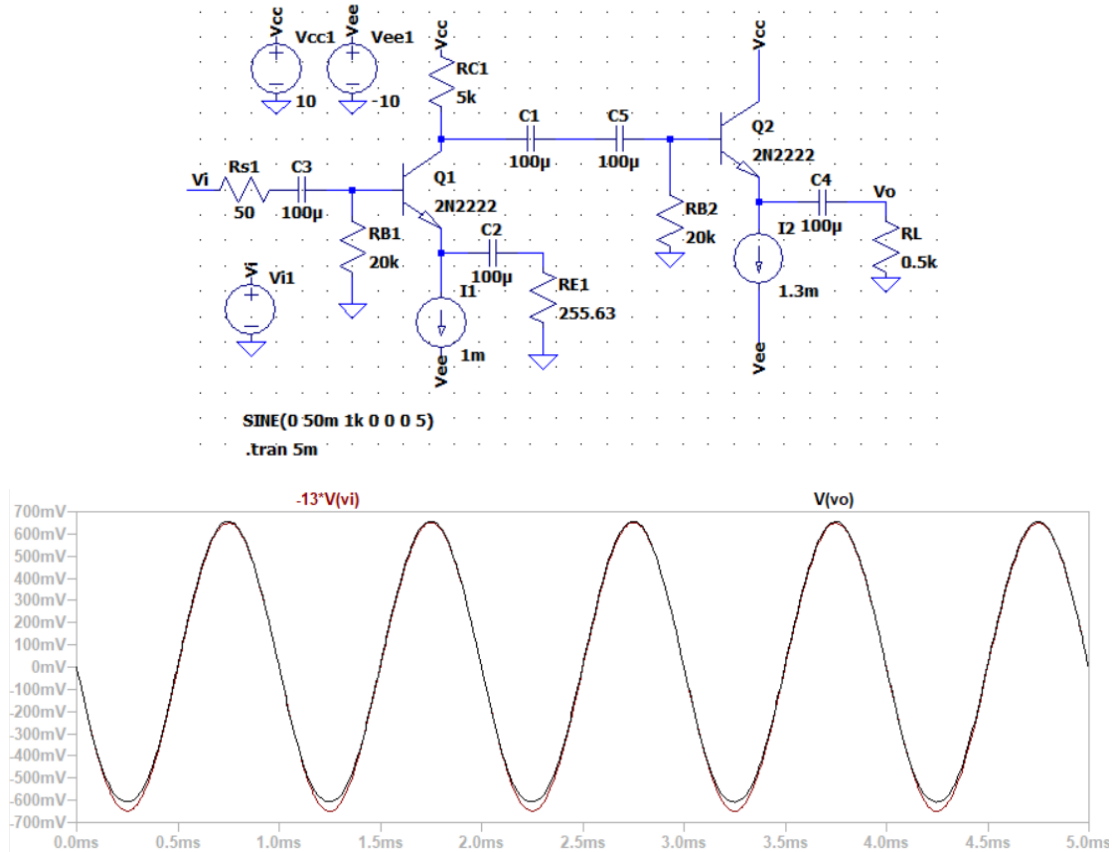


Figure 11. Simulation of the Modified Two-Stage CE-CC Amplifier

As shown in **Figure 12**, the maximum v_{be} of Q1 was measured to be

$v_{be-max} = (658.47 - 648.8)/2 = 4.48[mV]$, which is less than $5[mV]$. The measured amplitude of v_{be} of Q2 was not within $5[mV]$ because the magnitude of the input signal voltage of the CC amplifier was magnified to $13.65 \times 50[mV] = 682.5[mV]$. This would definitely result in a large variation of v_{be} .

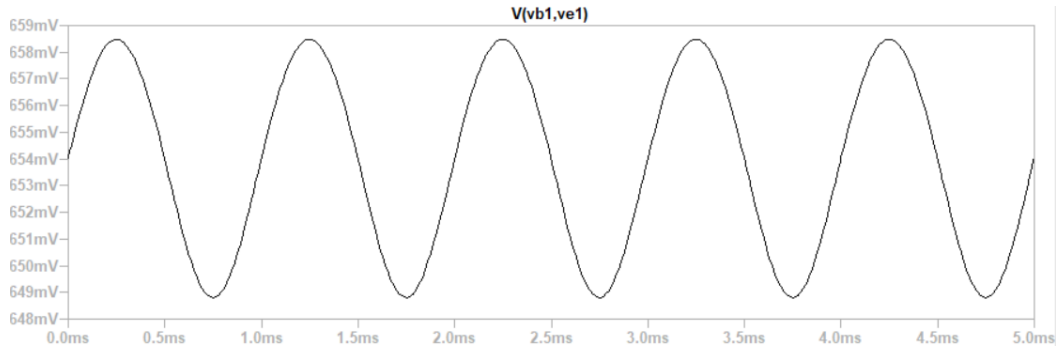


Figure 12. Measurement of v_{be} of Q1.

Figure 13 verified that the emitter current i_E for both transistors were above 0.

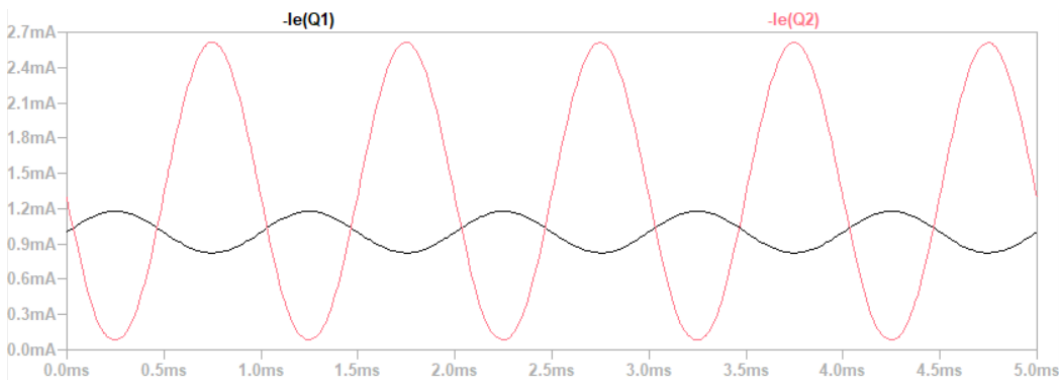


Figure 13. Measurement of i_E for both Transistors.

Figure 14 verified that v_{CE} for both transistors were above 0.2V. So, the two transistors stayed away from saturation.

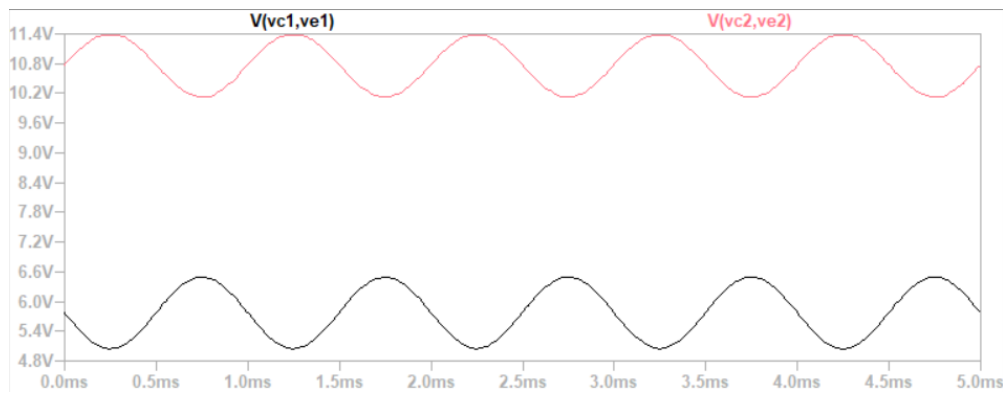


Figure 14. Measurement of v_{CE} for both Transistors.

When I_2 was set to 2[mA], the signal was distorted much less, as shown in **Figure 15**. In addition, v_{be} of Q2 was smaller though still greater than 5[mV].

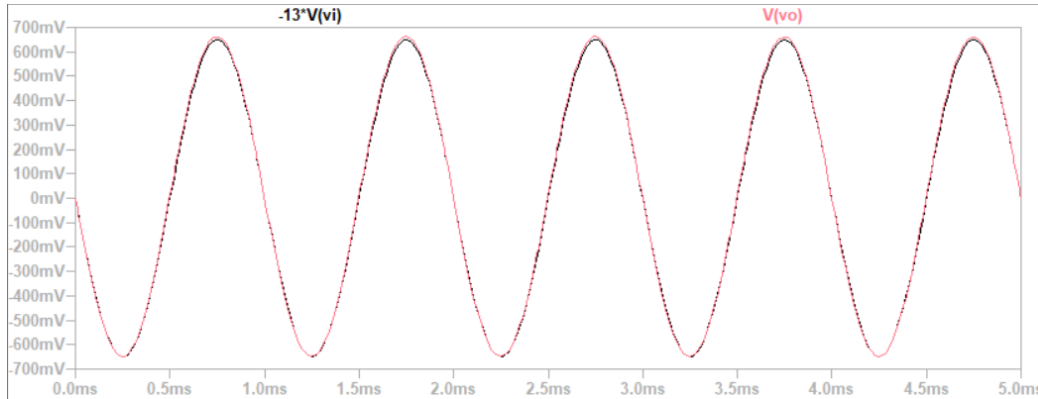


Figure 15. Verification of the Total Gain when $I_2 = 2[mA]$.

Conclusion

The design of the one-stage CE amplifier did not work with the given load resistance R_L . This was because of its large signal distortion, where v_{be} was substantially greater than 0.2[V]. The design of the two-stage CE-CC amplifier was successful. This showed that with a fixed gain, connecting a larger R_L to the CE amplifier decreases its v_{be} by demanding a larger R_E . A modification to the initial condition of I_2 was made to accommodate the BJT going into cut-off mode. After this modification, the output of the two-stage amplifier closely followed the expected result. The processes undertaken by this project has effectively enhanced our understanding of the CE and CC amplifiers.